

The Alberta Buck - Ethereum Implementation (DRAFT v0.92)

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A concrete Ethereum implementation of the Alberta Buck Architecture, in five contracts:

1. **BUCK_CREDIT** (ERC-721) – one NFT per insured asset, with deterministic depreciation and piecemeal client activation.
2. **IdentityRegistry** – per-address Pointcheval-Sanders credentials and ElGamal-encrypted identity points, verified on-chain via the BN254 precompiles. Trust anchor for every BUCK transfer.
3. **BUCK** (ERC-20) – identity-bound, single-slot per-account state, continuous demurrage, mutual-insurance-pool minting that draws coverage from BUCK_CREDIT NFTs cheapest-first.
4. **Notes** – privacy-preserving denomination on top of BUCK. Mint and spend via Groth16 SNARKs over a Tornado-style incremental Poseidon Merkle tree.
5. **BUCK_K** – the value-stabilization factor for credit-limit computation. Both a governance-set static variant and a full PID + Uniswap-V3-pool basket controller ship today; mint and burn drive the PID through `compute()`, and the controller is wired with priming, setpoint-shift (dS) compensation, and a long-gap `dTMax` clamp following the `ezpwd::pid` pattern.

This is the first implementation, prioritizing clarity over breadth of standards. Subsequent documents will examine ERC-1155, ERC-3643, and ERC-4626 alternatives for individual layers.

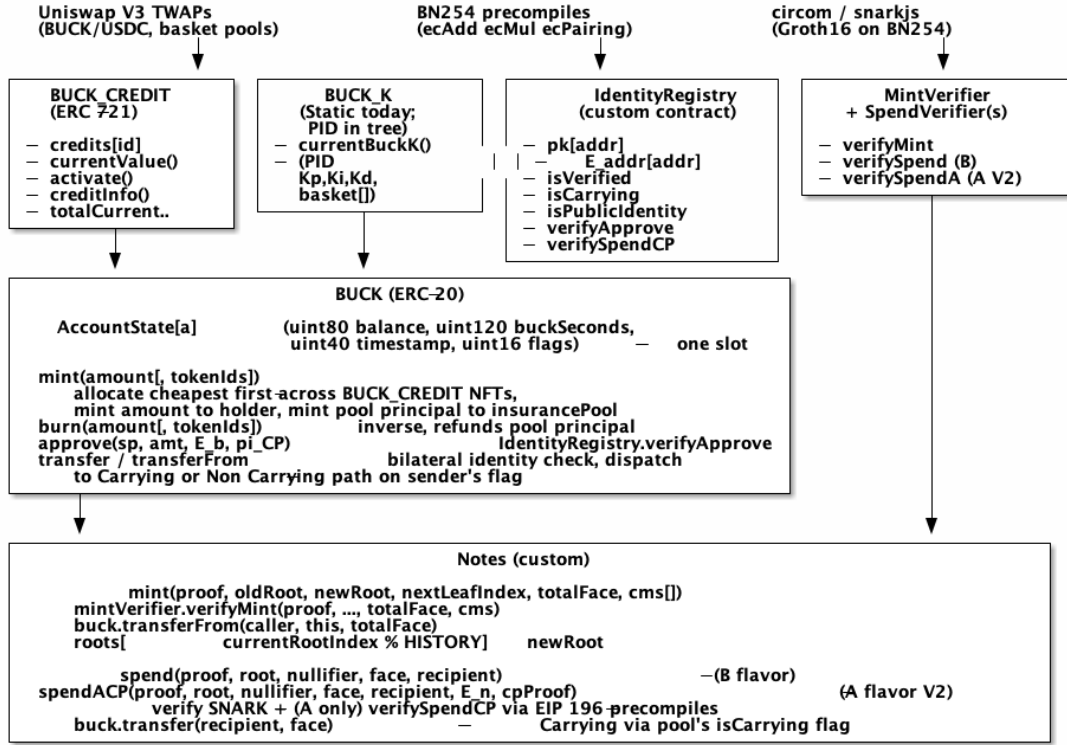
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1 Contract Architecture

The five contracts compose into a single transfer pipeline gated on identity. **IdentityRegistry** is the trust anchor: **mint**, **approve** and **transfer** all consult it before mutating state. Every account *and* contract that touches BUCK carries an identity binding – EOAs via **register**, contracts via **bindContract** – so bilateral identity-checked transfers always have a verifiable counterparty on both sides.



The flow is:

1. An **insurer** mints a BUCK_CREDIT NFT for a client's asset, setting face value, depreciation schedule, and **premiumRate** (annual basis points of activated coverage).
2. The **client** activates some or all of the credit via **BuckCredit.activate**, exposing the activated portion as collateral for **BUCK.mint**.
3. Before touching BUCK, the client **register=s** with **=IdentityRegistry**, posting a rerandomized PS credential and ElGamal-encrypted identity point. The registry verifies both via BN254 pairings (~235K gas, paid once per address).
4. **BUCK.mint(amount[, tokenId])** draws coverage cheapest-first across the client's NFTs, delivers exactly **amount** to the client, and mints a *mutual-insurance pool principal* (sized at 10x the annual premium on the activated coverage) to **insurancePool**.
5. To send BUCK, the sender **approves** with a Chaum-Pedersen receipt that re-encrypts the sender's identity under the spender's public key. **transfer / transferFrom** then bilateral identity-check both sides and emit **BuckTransferReceipt** carrying only ciphertext hashes.

6. To denominate as private notes, the client calls `Notes.mint` with a Groth16 proof binding the deposited `totalFace` to a batch of Poseidon commitments. Holders later call `Notes.spend` or `Notes.spendACP` with a fresh nullifier and a SNARK proving inclusion under a recent root.
7. `BUCK_K` ships in two forms today. The governance-set static variant is the simpler choice for early deployments where no BUCK price oracle exists yet. The PID + Uniswap-V3-pool variant reads its process variable (BUCK price) from a BUCK/USDT TWAP and its setpoint (basket cost) from a four-pool basket – XAUT/USDT, PAXG/USDC, cbBTC/USDC, WBTC/USDT, 25 % each – chosen for resilience against any single basis-currency or asset-token freeze. `Buck.mint` and `Buck.burn` call `BuckK.compute()`, so user activity drives and amortizes the PID; long quiet stretches are bounded by `dtMax`.

2 ERC Standard Choice for BUCK_CREDIT

ERC-721 was the obvious starting point and the choice that ships. Each credit is genuinely unique (specific asset, depreciation curve, insured value, premium schedule); `ERC721Enumerable` gives `BUCK.mint` the `tokenOfOwnerByIndex` it needs to walk a holder’s NFTs. ERC-3643 (T-REX) is a plausible future move if regulatory KYC becomes a hard requirement; ERC-1155 helps only if standardized credit *classes* emerge with shared depreciation parameters; ERC-4626 fits the `InsurancePool` but not individual credits. None of those are revisited below – the rest of the document specifies the shipped ERC-721 path.

3 BUCK_CREDIT: ERC-721 Insured Asset NFT

Each `BUCK_CREDIT` is an `ERC721Enumerable` token whose per-token `CreditParams` holds the insurer’s current offer and the client’s activation state. Three roles touch it: the **insurer** mints and updates, the **client** activates, `BUCK` reads the depreciated activated value to size mint allocations. All monetary fields are in 6-decimal BUCK-equivalent units (USDC-compatible).

3.1 Data Model

Fields group by mutability:

- **Immutable at mint:** `insurer` (the address allowed to update this token), `assetClass`, `createdAt`.
- **Insurer-mutable** via `updateCredit`: `faceValue`, `depreciationFloor`, `depType`, `depRate`, `depStartAt`, `premiumRate`, `lastUpdated`.
- **Client-mutable** via `activate`: `activatedValue` (capped at `faceValue`), `lastActivatedAt`.

`DepreciationType` is one of `NONE` (land, gold), `LINEAR` (constant annual reduction), or `DECLINING_BALANCE` (continuous-compounded percentage). Three events – `CreditCreated`, `CreditUpdated`, `CreditActivated` – record every state change.

3.2 Depreciation Models

`currentValue(tokenId)` is a pure view that branches on `depType`:

- `NONE`: returns the activated face value, unchanged.

- **LINEAR**: loss accrues at `depRate` bp/year on the depreciable portion (`faceValue - floor`), clamped at `depreciationFloor`.
- **DECLINING_BALANCE**: discrete whole-year compounding – the factor $(BP - \text{depRate})/BP$ is applied once per complete year via a loop capped at 100 years, then a linear interpolation bridges the trailing partial year. No transcendental approximations.

In every case the activated portion depreciates proportionally: `currentValue = depreciatedFace * activatedValue / faceValue`.

Asset	Type	Rate	Floor	30yr value (of \$100K)
Land	NONE	–	–	\$100,000
House (bldg)	LINEAR	200bp	20%	\$52,000
Vehicle	DECLINING_BALANCE	1500bp	5%	\$5,725
Farm equip	DECLINING_BALANCE	1000bp	10%	\$13,815
Gold in vault	NONE	–	–	\$100,000

A residential property would typically have two `BUCK_CREDITS`: one for the land (NONE) and one for the structure (LINEAR or DECLINING_BALANCE with a salvage floor).

3.3 Activation

The client activates piecemeal via `activate(tokenId, amount)` – gated on `ownerOf(tokenId) == msg.sender` and capped at `faceValue`. Activation just updates `activatedValue` and `lastActivatedAt`; the premium for newly activated coverage is collected by `BUCK.mint` when the client actually draws against it. The activation step exists so the client can *reserve* insurance against the asset without yet minting against it.

`totalCurrentValue(account)` walks `tokenOfOwnerByIndex` and sums `currentValue(tokenId)`. This is the single call `BUCK.mint` makes to size the credit limit. `creditInfo(tokenId)` returns (`faceValue`, `activatedValue`, `premiumRate`) in one tuple for the per-NFT walk inside `BUCK.mint`’s allocator.

3.4 Insurer Updates

`updateCredit(tokenId, newFaceValue, newDepreciationFloor, newDepType, newDepRate, newDepStartAt, newPremiumRate)` is gated on `msg.sender == credits[tokenId].insurer`. If the new face value is below the current activated amount the contract caps `activatedValue` downward in the same call. `assetClass` and `insurer` never change. Reductions can push a client’s outstanding `BUCK` above their credit limit; default insurance (held in `insurancePool`) covers the gap.

4 IdentityRegistry: On-Chain Trust Anchor

The `IdentityRegistry` is the binding between Ethereum addresses and the privacy-preserving identity machinery described in the companion documents *Identity with Anonymity*, *Identity Example – Data Flow*, and *Cryptographic Proofs*. Every `BUCK mint`, `approve` and `transfer` consults this registry; an unregistered address cannot hold or move `BUCKs`.

The registry stores per address: `pk[a]` (`G_1` identity public key), `E_addr[a]` (ElGamal ciphertext of the identity point `M`), `isVerified` (true after `PS+NIZK` pass on-chain or after `bindContract`), `isCarrying` (controls Buck’s transfer dispatch), and `isPublicIdentity` (true only for contracts

bound with the public-disclosure flag). Trusted-issuer PS public keys live in `trustedIssuers[issuerAddr]`, populated by governance. EOAs always self-register; contracts always `bindContract`.

4.1 Cryptographic Surface

Three on-chain verifications, all reducible to BN254 precompile calls:

1. **PS signature at registration:** $e(\text{sigma}'_1, X + m * Y) = e(\text{sigma}'_2, g_2)$, with m reconstructed linearly via the binding NIZK in step 2. $\sim 147K$ gas.
2. **Schnorr-family NIZK** binding the rerandomized PS signature to the ElGamal ciphertext. Three response scalars, Fiat-Shamir bound to `(sigma', E_new, pk_new, registrant_address)`. $\sim 30K$ gas on top of step 1. Total registration $\sim 235K$, paid once per address.
3. **Chaum-Pedersen NIZK at approve:** proves the recipient's identity ciphertext re-encrypts the same M registered in `E_alice`. Fiat-Shamir binds `msg.sender`, `spender`, and `chainid`. $\sim 29K$ gas per counterparty pair.
4. **Chaum-Pedersen DLEQ at A-flavor spend** (`verifySpendCP`): proves the spender holds a single `sk_dep` that opens both the note's `E_n` and the registered `E_reg`. $\sim 36K$ gas via EIP-196.

4.2 Solidity Surface

`src/IdentityRegistry.sol` uses the BN254 helper for all curve math:

```
struct G1Point { uint256 X; uint256 Y; }
struct G2Point { uint256[2] X; uint256[2] Y; }

struct ElGamalCT { G1Point R; G1Point C; } // (R, C) = (r*G, r*pk + M)
struct PSSig { G1Point sigma_1; G1Point sigma_2; }
struct PSPubKey { G2Point X; G2Point Y; }

struct RegistrationProof { uint256 e, s_m, s_r; G1Point A_ps, T_C, T_R; }
struct CPProof { uint256 e, s1, s2; G1Point T1, T2, T3; }
struct SpendCPProof { uint256 e, s; G1Point T1, T2; }

function register(address issuer, G1Point pk, ElGamalCT E,
    PSSig sigma, RegistrationProof proof);
function bindContract(address target, G1Point pk, ElGamalCT E,
    bool isPublicIdentity_, bool isCarrying);
function trustIssuer(address issuer, PSPubKey ipk); // governance
function verifyApprove(address sender, address spender,
    ElGamalCT E_bob, CPProof pi) view returns (bool);
function verifySpendCP(address spender, address recipient,
    ElGamalCT E_n, SpendCPProof cpProof) view returns (bool);
function markApproved(address spender); // Buck-only
function setBuck(address buck); // governance, one-shot
```

A few non-obvious points:

- `verifyApprove` reads `E_alice` from registry storage so the caller cannot substitute a fake credential.
- `bindContract` is permissioned by deployment pattern, not governance: any caller may bind any yet-unbound deployed contract. The intended discipline is operator OPSEC – either the operator is the deployer and uses `BuckAwareDeployer.deployAndBind` for atomic deploy+bind, or the operator races to bind a pre-existing contract before anyone else. Once bound, immutable.
- `markApproved` (called by Buck inside `approve`) freezes the spender’s `isCarrying` flag the first time anyone successfully approves them, so the carrying flavor a sender consents to cannot be retroactively flipped.
- Revocation does not un-register existing accounts; they were verified once and are never re-verified.

4.3 Public-Identity Contracts

DeFi infrastructure – AMM pools, vaults, oracles, the Notes pool – can’t generate Chaum-Pedersen proofs at `approve` time because the contract has no private key. Such contracts bind via `bindContract(target, pk, E, isPublicIdentity_=true, isCarrying=true)`. Counterparties of a Public-Identity contract still call the identity-bound `approve` (the always-CP rule applies regardless of spender flavor), but the receipt-fragment check on the contract side is satisfied by the deterministic `_identityHash` rather than a per-counterparty CP receipt. Public-Identity contracts forfeit content privacy: their identity M is openable on subpoena via the operator’s off-chain attestation. Encrypted-Identity contract binding (`isPublicIdentity_=false`) is supported by the surface but not yet used in production.

4.4 Issuer Trust

`trustedIssuers` is the only governance-controlled trust anchor. Adding an issuer admits a new origin of registrable identities; removing one stops new registrations under that issuer. Existing registrations are unaffected – they were verified at registration and never re-verified. Epoch-based credential renewal, key revocation, and issuer rotation are described in Identity: Epoch-Based Credential Renewal.

5 BUCK: ERC-20 Token

BUCK is an identity-bound ERC-20 with mutual-insurance-pool minting and continuous demurrage. All transfer-path state for an account fits in one storage slot, so a transfer is one `SSTORE` per side. `mint(N)` delivers exactly N BUCK to the holder and simultaneously mints a *pool principal* to `insurancePool` sized so that, at the insurer’s assumed annual ROI, the principal’s investment yield exactly covers the annual premium on the activated coverage. `burn(N)` is the inverse. Every transfer routes through the `IdentityRegistry` and emits `BuckTransferReceipt(from, to, amount, fromHash, toHash)` – ciphertext hashes, no plaintext identity on chain.

5.1 Storage Layout

The transfer-path state for an account is one slot:

```

struct AccountState {
    BuckQty      balance;          // uint80 underlying; ~1.21e24 cap (1.21e18 BUCK at 6 decimals)
    BuckSeconds  buckSeconds;      // uint120 underlying; demurrage IOU integral
    uint40       timestamp;        // last crystallisation (safe past year 36800)
    uint16       flags;            // reserved for future per-account flags
}
mapping(address => AccountState) internal _state;

```

Everything outside the hot path (allowances, mint-side limits, per-NFT outstanding allocations, approve-time receipt fragments) lives in separate maps because they're touched per-mint, not per transfer. BUCK implements IERC20 and IERC20Metadata directly – no OpenZeppelin ERC20 inheritance, so the slot layout is explicit and the transfer paths are linear (no `_update` override re-entry). Decimals are 6, matching USDC.

5.2 mint(N) and burn(N)

`mint(uint256 amount)` and `mint(uint256 amount, uint256[] calldata tokenIds)` are the two overloads. The no-arg form runs an internal cheapest-first selector (insertion sort over the caller's NFT list – typically a handful) to minimise premium cost; the second form takes a caller-supplied order an off-chain optimizer pre-computes against effective rate, depreciation, or coverage midpoint.

`burn(uint256 amount)` mirrors the overload pair, but the default selector walks the holder's NFTs **most-expensive-first** – the dearest insurance is released first, returning the largest pool principal per BUCK burned and freeing expensive coverage capacity for re-use. The asymmetry is not arbitrageable: per-NFT inversion is symmetric (same `denom` on both sides), so a mint-burn round-trip on the same NFT restores its `mintsBacked` exactly.

The *mutual-insurance pool model*: each NFT carries an annual `premiumRate` in basis points. Drawing `take` units of coverage from one NFT generates an annual premium of `take * premiumRate / BP`. The pool deposit sized to cover that premium at a 10% assumed ROI is `take * premiumRate * POOL_ROI_INV / BP` where `POOL_ROI_INV = 10`. The insurer keeps any return above 10% as profit. Per-NFT inversion – given a target net delivery `delivery` and rate `r`:

```

denom      = BP - r * POOL_ROI_INV          (must be > 0; max effective rate ~1000bp)
take       = ceil(delivery * BP / denom)
principal  = take - delivery

```

The allocator walks the NFT list in order, computes per-NFT `netCap = avail * denom / BP`, and either fully consumes the NFT (delivery contribution = `netCap`, principal = `take - netCap`) or partially consumes it via the closed-form inversion to land exactly on `remaining`.

Per-NFT bookkeeping: `mintsBacked[tokenId]` tracks the cumulative outstanding allocation against each NFT, capped at `activatedValue`. Burn decrements; mint increments. A mint-burn round-trip on the same NFT order is rate-neutral and not arbitrageable – both sides hit the same `denom`.

The shipped mint sequence:

1. `isVerified(msg.sender)` guard.
2. Read `buckCredit.totalCurrentValue(sender) * buckK.compute() / 1e18`, ratchet `storedLimit[sender]` upward only. `compute()` advances the PID if `dT` has elapsed (oracle reads + state update) or returns the cached value otherwise – mint and burn together amortize PID work across user activity.

3. Early gate: `_state[sender].balance (raw) + amount <= storedLimit` – uses raw balance so that accrued demurrage fees do not silently inflate the apparent headroom; `amount` is a lower bound on total coverage drawn (since pool principal is non-negative).
4. `_allocateMint(amount, tokenIds)` walks the NFTs cheapest-first (or in caller order), writes `mintsBacked`, returns `(totalCoverage, poolPrincipal)`.
5. Tight gate: `_state[sender].balance (raw) + totalCoverage <= storedLimit`.
6. `_accrueJubilee()`, crystallise the holder, credit `amount` to the holder; if `poolPrincipal > 0`, crystallise `insurancePool` and credit `poolPrincipal` to it. `_totalSupply + amount + poolPrincipal =`.
7. `Emit Transfer(0, sender, amount), Transfer(0, insurancePool, poolPrincipal), Minted(sender, totalCoverage, poolPrincipal, creditValue, buckKValue, limit)`.

`burn(amount[, tokenIds])` mirrors the math but with the most-expensive-first default selector: unwind dearest NFT first, decrement `mintsBacked`, burn `amount` from the holder and `poolRefund` from `insurancePool`, drop `totalSupply` by `amount + poolRefund`. `quoteMint` and `quoteBurn` are read-only views that return `(totalCoverage, poolPrincipal)` or `(totalUnwind, poolRefund)` for the same allocator math, so an off-chain optimizer can size required limits and refunds before submitting.

5.3 Identity-Bound approve()

`approve(spender, amount, E_bob, pi_CP)` is mandatory – the parameterless ERC-20 `approve(address, uint256)` reverts with BUCK: `use identity-bound approve`. Both `sender` and `spender` must be `isVerified`. The CP proof is required regardless of the spender’s identity flavor (Public-Identity or Encrypted): the always-CP rule produces a uniform per-pair audit record openable on subpoena, and engages the spender’s BN254 identity key for forward security under future identity rotation. Full rationale in *Why approve Is Always CP-Bound*.

On success, `_receiptFragments[sender][spender] = keccak256(E_bob)`, the standard ERC-20 allowance is written, and `markApproved(spender)` freezes the spender’s `isCarrying` flag in the registry (so the carrying flavor consented to cannot retroactively flip). Per-pair cost is ~97K gas (29K CP verify + 46K allowance + 22K receipt SSTORE), paid once. Subsequent transfers between the same pair add only the bilateral check and the receipt event (~5K above an ERC-20-baseline transfer).

5.4 transfer() and transferFrom()

Both route through `_identityCheckedTransfer`, which enforces *mutual decryptability*: every private (non-public) party must hold a per-pair CP receipt fragment for its counterparty, established before the transfer by the 4-arg identity-bound `approve`. Only public-identity contracts may use the deterministic `_identityHash` fallback.

- Requires `isVerified(from)` and `isVerified(to)`.
- For the to side (`_receiptFragments[from][to]`): if zero, requires `=isPublicIdentity(from)` — reverts with BUCK: `sender must identity-approve recipient` otherwise. Falls back to `_identityHash(to)` when `from` is public.

- For the `from` side (`_receiptFragments[to][from]`): if zero, requires `=isPublicIdentity(to)` — reverts with BUCK: recipient must identity-approve sender otherwise. Falls back to `_identityHash(from)` when `to` is public.
- Both guards pass only when *both* parties are public-identity contracts. EOA-to-EOA transfers require bilateral CP-approve; EOA-to-Public and Public-to-EOA require the private party to CP-approve the public one.

After identity checks, transfer dispatch is automatic on the sender’s `isCarrying` flag: Carrying senders (AMM pools, the Notes pool, the Jubilee fund) take the `_carryingTransfer` path that crystallises both sides through now and apportions the sender’s live `buckSeconds` proportionally to the value transferred; Non-Carrying senders take `_nonCarryingTransfer` which just shifts raw balance after a spendable check on `balanceOf(from) = balance - feeOwing`. `BuckTransferReceipt(from, to, amount, fromHash, toHash)` is emitted; `_identityHash(a) = keccak256(abi.encode(pk, E_addr))` is deterministic per account.

5.5 Demurrage and Jubilee

BUCK accrues continuous demurrage to fund a perpetual Jubilee. Each account’s `buckSeconds` field is the cumulative integral of $(\text{raw balance} \times \text{dt})$ crystallised at `timestamp`; the live fee owed is

$$\text{feeOwing}(a) = (\text{buckSeconds} + \text{balance} * (\text{now} - \text{timestamp})) * \text{BASE_RATE_PER_SEC} / \text{SCALE}$$

with `BASE_RATE_PER_YEAR = 0.02` (2%/year) and `SCALE = 1e27`. Visible balance:

- **Non-Carrying** (default for EOAs): `balanceOf(a) = max(0, balance - feeOwing(a))`. The locked fee stays inside the account’s own slot; spendable shrinks over idle periods.
- **Carrying** (AMM pools, Notes, Jubilee, etc.): `balanceOf(a) = balance`. The account never loses visible balance to demurrage; instead, on every outflow the value-weighted age is folded into the recipient’s `buckSeconds` so the locked debt rides with the BUCKs.

`_crystallize(a)` folds the time rectangle into `buckSeconds` and bumps `timestamp`; it never changes balance. Carrying transfer crystallises both sides through now, apportions `liveBs * value / raw` of the sender’s live integral to the recipient (where `liveBs = buckSeconds + raw * (now - timestamp)`), and settles each side in one SSTORE.

The Jubilee is a normal account at `address(this)`, treated as Carrying. On every mint or burn, `_accrueJubilee` writes

$$\text{_state[address(this)].balance} += \text{totalSupply} * \text{BASE_RATE_PER_SEC} * \text{elapsed} / \text{SCALE}$$

directly into Jubilee’s slot. `totalSupply` is **not** mutated – demurrage is internal redistribution, not minting. The exact invariant:

$$\text{sum_a rawBalanceOf}(a) == \text{totalSupply} + \text{jubileeActual}()$$

Jubilee is the system-wide fee escrow. Every BUCK has exactly one fee owner: non-Carrying accounts lock it silently inside their raw balance (`balanceOf = raw - feeOwing`); Carrying accounts (Notes pool, AMM pools, Jubilee itself) show `balanceOf = raw`, so their fee portion sits in Jubilee instead. The `_accrueJubilee` call credits Jubilee at the full `totalSupply` rate, covering Carrying

holdings too. When Carrying BUCKs flow to a non-Carrying account via `_carryingTransfer`, the recipient absorbs `liveBs * value / raw` as extra `buckSeconds`, where `liveBs` is the sender's full live integral (crystallised history plus the current elapsed rectangle). Jubilee pre-accrued the full fee at the `totalSupply` rate; it now debits in exact proportion to the BUCKs transferred, with both sides settled through the current block. `JubileeAccrued(delta, newBalance)` surfaces every accrual.

The Jubilee is the sink that funds default coverage, perpetual-ground-rent disbursement, or a periodic balance reset – all governance-driven on-chain disbursement paths land on top of this single accumulating slot.

6 BUCK_K: Value Stabilization Controller

BUCK_K supplies the credit-limit multiplier `BUCK.mint` reads to size against `totalCurrentValue`. The `IBuckK` interface is two functions – `currentBuckK()` external view returns `(uint256)` and `compute()` external returns `(uint256)` – so BUCK is invariant to which implementation is wired in. `BUCK.mint` and `BUCK.burn` both call `compute()` so user activity drives the controller and amortises any PID work; the function is otherwise permissionless, keepers and individual users can poke it directly during quiet stretches. Two implementations ship today: `BuckKControllerStatic` (governance-set constant) and `BuckKController` (full PID + Uniswap-V3-pool basket).

6.1 Static Variant

`BuckKControllerStatic` holds one `uint256 buckK` (18 decimals, `1e18` = neutral). Functions: `currentBuckK()`, `compute()` (returns the stored value, non-view to match the `IBuckK` signature), `setBuckK(uint256)` (governance-only), `transferGovernance(address)`. This is the simpler choice for early deployments where no BUCK price oracle exists yet, exercising the identity, demurrage, and Notes layers while `BUCK_K` is held constant.

6.2 PID Variant

`BuckKController` follows the `ezpwd::pid` controller layout, ported to 18-decimal Solidity fixed-point.

PID Term	Source	Meaning
Process Variable	BUCK/USDT Uniswap V3 TWAP	What a BUCK actually trades for
Setpoint	Commodity-basket TWAP sum	What a BUCK should be worth
Error	Process - Setpoint	Positive: BUCK overvalued; negative: undervalued
Output (<code>BUCK_K</code>)	$1e18 + (P \cdot K_p + I \cdot K_i + D \cdot K_d) / 1e18$	Credit-limit multiplier (1.0 = neutral)

Sign convention (all shipped gains are **positive**):

- **BUCK below basket (inflation):** `error < 0` -> `BUCK_K decreases`. Credit contracts; holders running near their limit must burn BUCK to stay compliant; supply shrinks; BUCK price recovers toward basket from below.
- **BUCK above basket (deflation):** `error > 0` -> `BUCK_K increases`. Credit expands; holders can mint additional BUCK against their NFTs and sell it into the pool at the premium; supply grows; BUCK price recovers toward basket from above.

Two canary tests in `test/BuckKController.t.sol` guard this direction (`test_sign_convention_inflation_co` and `test_sign_convention_deflation_expands_credit`); any sign flip in the controller surface fails them within one PID cycle. See alberta-buck-ethereum-example.org for the bootstrap, equilibrium, and arbitrage examples that exercise the loop end-to-end.

6.2.1 Basket and BUCK Price Oracles

The controller reads basket components from two parallel sources, both summed into the basket cost: a Chainlink-feed array (`BasketComponent { feed, weight, feedDecimals }`, used in tests against mock aggregators) and a Uniswap-V3-pool array (`BasketPool { pool, baseToken, quoteToken, weight, baseDecimals, quoteDecimals, twapInterval }`). `addBasketComponent` and `addBasketPool` are governance-only. The shipped target basket is V3-pool-sourced:

Pool	Base	Quote	Weight
XAUT / USDT	XAUT (6d)	USDT	25 %
PAXG / USDC	PAXG (18d)	USDC	25 %
cbBTC / USDC	cbBTC (8d)	USDC	25 %
WBTC / USDT	WBTC (8d)	USDT	25 %

Two gold tokens and two BTC tokens, paired against two stablecoins, gives the basket resilience against any single asset-token or basis-currency freeze. `setBuckPriceOracle(pool, buckToken, quoteToken, quoteDecimals, twapInterval)` configures the BUCK price oracle (typically BUCK/USDT).

Pool reads route through `UniswapV3OracleLib (src/lib/)` – a 0.8-compatible port of the `OracleLibrary /=TickMath= /=FullMath=` helpers we need. The vendored `lib/v3-periphery OracleLibrary` is pinned to `<0.8.0` and can't be imported directly under `0.8.20+`. When a pool's `twapInterval > 0` the controller calls `consult(pool, secondsAgo)` for the time-weighted mean tick; when zero it reads `slot0().tick` directly (test-only).

6.2.2 Priming and dS Compensation

The first `compute()` after deployment captures references and returns the cached `buckK` without running PID math:

```
prime cycle: P = current_error; lastBasketCost = current_basket; primed = true;
              return buckK
```

This avoids the spurious first-cycle derivative spike that arose from `P_prev = 0` (default storage) combined with a non-zero initial error. All subsequent cycles compute

```
dS = basketCost - lastBasketCost
D  = (error - P_prev - dS) * UNIT / dt
```

where `dS` captures setpoint movement (gold spike, governance basket re-weight, oracle update). Subtracting `dS` from `(error - P_prev)` strips out target movement so derivative reflects only BUCK's drift relative to its (moving) target. Without `dS`, a basket-side jump would otherwise fire `Kd * basketJump / dt` as if BUCK had moved, slamming `BUCK_K` toward a rail.

6.2.3 dT, dTMax, and Pokability

Three time constants govern when and how aggressively the controller responds:

- **dT (minimum cycle interval):** `compute()` short-circuits if `now - lastUpdate < dT` and returns cached `buckK` as a one-SLOAD operation. This is the amortisation knob – multiple mints in a single dT window share one PID cycle’s gas cost. 60 s is reasonable for a chain with ~12 s blocks (one PID cycle per ~5 blocks).
- **dTMax (long-gap clamp):** when `now - lastUpdate > dTMax`, the cycle integrates as if only dTMax seconds had passed. Real `lastUpdate` still advances to `now`, so subsequent cycles measure forward correctly. Default `type(uint256).max` (no clamp); governance opts in via `setDTMax`. Typical value 1 h: a 24-h silence then integrates as one 1-h step rather than slamming the integral with 24 h of accumulated error.
- **Pokability:** `compute()` is permissionless and side-effect-only-when-due. In normal operation `Buck.mint / Buck.burn` drive it; in long quiet stretches a keeper can poke it directly to keep the integral fresh, or rely on `dTMax` to bound the eventual catch-up step.

The TWAP window itself (currently 600 s on every pool) is the manipulation-resistance knob. A flash-loan-shape attack that drives BUCK to \$0.50 in one block and unwinds the next leaves spot collapsed but TWAP unmoved – the test suite verifies that during such an attack the TWAP read sits at ~\$1.000002 while spot reads \$0.500.

6.2.4 Future: Amortizing PID Work via a State Machine

Today’s `compute()` reads all five pools and runs the full P/I/D math in a single call. At ~5 V3 pool reads + a few mults this is cheap (~300-500 K gas including a state write). If a richer PID grew expensive enough that mint-time amortization became painful, the cycle work could be split into a state machine over multiple calls – each transfer advancing one micro-step of (read pool i, accumulate, finalize). The tradeoff is that pool reads stretch over real time and so the basket sample becomes incoherent unless every input is itself TWAP’d over a window covering the state-machine duration; today’s full-cycle compute wins for that reason. Revisit when oracle reads grow or the basket expands materially.

6.2.5 Tuning

Gain	Effect of increasing	Risk
Kp	Faster response to deviations	Oscillation, over-correction
Ki	Eliminates steady-state error	Integral windup, slow recovery
Kd	Damps oscillation, anticipates trends	Noise amplification

Initial gains should be conservative (high Kd relative to Kp, low Ki). dT acts as a coarse damper, dTMax as a backstop. Priming and dS compensation make non-zero Kd safe at small dT: pre-priming, every fresh deploy with Kd > 0 had a permanent derivative contribution because `P_prev` was stuck at 0 – post-priming, `P_prev` captures the actual initial error and the derivative dies off as soon as the system reaches a new equilibrium.

A future enhancement is on-chain gain scheduling: maintain a rolling variance of the error signal and rotate between conservative and responsive gain sets at regime boundaries. This maps to the `ezpwd::pid::reset()` hot-swap pattern in the C++ reference.

7 Notes: Privacy-Preserving Denomination

BUCK transfers carry a public (`from`, `to`, `amount`) quartet. Notes lift values, flavours, and identity bindings off the public ledger by escrowing BUCK in a commitment pool: holders post Poseidon-hash commitments via a Groth16 mint, then redeem any commitment later via a second Groth16 spend with a fresh nullifier. The contract’s public state is just a list of opaque 32-byte commitments, a 30-slot recent-roots ring buffer, a nullifier set, and a running face-value total. Full design in Notes and Proofs; this section documents the Ethereum surface only.

7.1 Mint Circuit (Batch, In-Circuit Merkle Insertion)

`circuits/mint_batch.circom` is a parameterised batch-mint circuit. Each pinned size N (specialised at $N \in \{1, 2, 4, 8, 16, 32\}$, with the smaller sizes useful for unit tests and small wallet flows) produces its own `MintBatchN*Groth16Verifier.sol`. The shipped Notes contract holds a single `mintVerifier` adapter; per- N dispatch lives inside the adapter (a single `IMintVerifier` chooses the right pinned verifier by `cms.length`). Public-signals arity is 5 regardless of N :

Signal	Description
<code>oldRoot</code>	Live note Merkle root the prover read from chain
<code>newRoot</code>	Root after folding <code>cm[0..N-1]</code> starting at <code>nextLeafIndex</code>
<code>nextLeafIndex</code>	Live tree size the prover read from chain
<code>totalFace</code>	Sum of note values, range-bounded to 128 bits
<code>cmBatchHash</code>	<code>keccak256(abi.encodePacked(cm[0..N-1]))</code> – ties calldata to the SNARK

Per leaf, the circuit enforces a Poseidon-5 commitment opening `cm_i === Poseidon(flavor_i, v_i, rho_i, idHash_i, predicate_i)`, `Num2Bits(128)` range bounds on `v[i]` and `totalFace` (closing field-overflow attacks), one sum check `totalFace === sum v[i]`, and an $N \times \text{TREE_DEPTH}$ Poseidon-2 dual Merkle walk: `priorWalk` consumes `ZERO_VALUE` at the insertion slot to attest `oldRoot`, `postWalk` consumes `cm[i]` to advance `rollingRoot` to `newRoot`. Both walks share prover-supplied siblings. A keccak hash check binds `cmBatchHash` to the per-leaf commitments. $\sim 6\text{K}$ R1CS per leaf ($\sim 1.7\text{K}$ opening + $\sim 4.3\text{K}$ insertion).

Verifier bytecode sizes: 14.5 KB at $N=16$, 20.8 KB at $N=32$ – both under the EIP-170 24 KB ceiling. Per-mint gas: 461K at $N=1$ (constant $\sim 216\text{K}$ Groth16 verifier dominates), 624K at $N=16$ ($\sim 39\text{K}/\text{leaf}$ – the shipped sweet-spot), 1.12M at $N=32$. Asymptotic per-leaf cost is $\sim 31\text{K}$. Promotion to $N \geq 64$ needs one of the three escape hatches in `alberta-buck-notes-rollup-mint.org` (calldata-IC hash-pinned, sharded SSTORE2, plain SLOAD), with calldata-IC the natural fit for the Holochain wallet.

The trusted setup in `scripts/snark/setup.sh` uses a fixed dev-only entropy string. Production would run a multi-party ceremony per pinned N ; the script is the scaffolding.

7.2 Spend Circuits (B-flavor, A-flavor V2)

`circuits/spend.circom` proves redemption of a single B-shape note under an on-chain Merkle root. Public inputs [`noteRoot`, `nullifier`, `face`, `recipient`, `chainId`]; private witness (`flavor`, `v`, `rho`, `idHash`, `predicate`, `pathElements[20]`, `pathIndices[20]`). Constraint groups: `Num2Bits(128)` on `face` and `v`, `cm === Poseidon-5(flavor, v, rho, idHash, predicate)`, Tornado-style depth-20 Merkle inclusion under `noteRoot`, B-tag nullifier `Poseidon-3(rho, idHash, 4242)`, and value conservation `face * 1 === v * 1` with a quadratic anchor. $\sim 12,000$ R1CS. `SpendGroth16Verifier` takes the 5-element public-signals tuple; `SpendVerifierAdapter` presents the narrow `ISpendVerifier`.

`circuits/spend_a.circom` V2 is the A-flavor (identity-bound) sibling. Public inputs grow to `[noteRoot, nullifier, face, recipient, chainId, E_n.R.x, E_n.R.y, E_n.C.x, E_n.C.y]` (9 total). Adds the A-tag nullifier `Poseidon-3(rho, idHash, 4243)`, a flavor-in-`{1,2}` gate, and an in-circuit `Poseidon-8` binding gate `Poseidon-8(E_n, issuerData) === idHash` (~265 R1CS) that pins the publicly-revealed `E_n` to the leaf at mint time. Total ~13,300 R1CS.

The cryptographic identity gate – proving the spender holds a single `sk_dep` that opens both `E_n` and the registered `E_reg` – lives in `IdentityRegistry.verifySpendCP` rather than the circuit. In-circuit BN254 G1 arithmetic over the BN254 scalar field would have cost +250K R1CS per group operation; off-chain CP-DLEQ verified via EIP-196 precompiles costs ~36K gas instead. `Notes.spendACP` calls the Groth16 verifier and `verifySpendCP` atomically; both must succeed. The DLEQ is a 4-element Sigma proof:

Prover picks `t` in `Fr`, computes `T1 = t*G`, `T2 = t*(R_reg - R_n)`. Challenge: `e = H(R_n, R_reg, C_n, C_reg, pk_dep, T1, T2, recipient, chainid)`. Response: `s = t + e*sk_dep`. Proof: `(e, s, T1, T2)` – 4 field elements, ~36K gas to verify on-chain.

The invariant the two layers together enforce: A-note loss-recovery via identity re-issuance is impossible by design. `sk_rec` must be backed up like a hardware-wallet seed; loss is terminal. Companion documents `Notes` and `Notes Flow` document the wallet-side implications.

7.3 Verifier Interfaces

`Notes` is decoupled from the SNARK toolchain via narrow interfaces:

```
interface IMintVerifier {
    function verifyMint(bytes proof, uint256 oldRoot, uint256 newRoot,
        uint256 nextLeafIndex, uint256 totalFace,
        uint256[] calldata cms) external view returns (bool);
}
interface ISpendVerifier {
    function verifySpend(bytes proof, uint256 root, uint256 nullifier,
        uint256 face, address recipient, uint256 chainId)
        external view returns (bool);
}
interface ISpendAVerifier {
    function verifySpendA(bytes proof, uint256 root, uint256 nullifier,
        uint256 face, address recipient, uint256 chainId,
        uint256 enRx, uint256 enRy, uint256 enCx, uint256 enCy)
        external view returns (bool);
}
```

Stub implementations (`StubMintVerifier` / `StubSpendVerifier`) ship for unit tests that exercise state transitions unrelated to the SNARK. The production `Adapter` contracts decode the proof as `(uint256[2], uint256[2][2], uint256[2])`, build the public-signals tuple, and forward to the snarkjs-generated Groth16 verifier. `MintVerifier` is a singleton `IMintVerifier` adapter that internally dispatches by `cms.length` to the correct pinned `MintBatchN*Groth16Verifier`; spend verifiers are likewise singletons. Swapping any of them is a single governance call.

7.4 Notes Contract

`src/Notes.sol` is constructed with (`buck`, `mintVerifier`, `spendVerifier`, `governance`). The A-flavor spend verifier and the identity registry are wired post-construction via governance setters (`setSpendAVerifier`, `setIdentityRegistry`); until both are set, `spendACP` reverts with `Notes: A-spend disabled or Notes: identity registry not set`. Its state is intentionally lean – per-leaf insertion math now lives in the SNARK:

- **Verifier handles:** singleton `mintVerifier` (the adapter that dispatches by batch size), singletons `spendVerifier` for B-spend and `spendAVerifier` for A-spend, plus the `identityRegistry` used by `spendACP` for `verifySpendCP`. All swappable by governance.
- **Audit / double-spend state:** `mapping(uint256 => bool) nullifiers, uint256 noteFaceSum`.
- **Tree state:** `uint32 nextLeafIndex` and `uint256[ROOT_HISTORY_SIZE=30] roots` with `uint8 currentRootIndex`. `filledSubtrees`, `zeros[]`, and per-cm uniqueness checks are gone – duplicate openings are economically self-punishing via the deterministic nullifier (one spend nullifies every copy of the same opening; the duplicator forfeits their mint).

`mint(proof, oldRoot, newRoot, nextLeafIndex, totalFace, cms)` asserts `cms.length > 0`, checks `oldRoot == roots[currentRootIndex]` and `nextLeafIndex == self.nextLeafIndex` (stale-state guards), field-bounds each `cms[i]` against `FIELD_R`, dispatches to `mintVerifier` (which forwards to the per-N pinned Groth16 verifier internally), pulls `totalFace` BUCK via `transferFrom(caller, this, totalFace)`, writes `newRoot` into the ring buffer at `(currentRootIndex + 1) % ROOT_HISTORY_SIZE`, advances `nextLeafIndex` by `cms.length`, bumps `noteFaceSum` by `totalFace`, and emits `Minted(issuer, totalFace, startIndex, count, newRoot)`. Per-leaf Appended events are gone; the commitments live in calldata for off-chain replayers.

`spend(proof, root, nullifier, face, recipient)` (B-flavor) requires `recipient != 0`, `face > 0`, `isAcceptedRoot(root)` (root in the recent-roots window), and `!nullifiers[nullifier]`. Forwards to `spendVerifier`, marks the nullifier burned, decrements `noteFaceSum`, and calls `buck.transfer(recipient, face)`. Because the Notes pool is bound under the Carrying flag, `buck.transfer` dispatches to the carrying path automatically – the recipient absorbs the pool’s average demurrage age via `buckSeconds`, preserving the SNARK’s `face === v` invariant at the ERC-20 boundary. `Spent(nullifier, face, recipient)` closes the spend.

`spendACP(proof, root, nullifier, face, recipient, E_n, cpProof)` (A-flavor) is identical except it also calls `identityRegistry.verifySpendCP(msg.sender, recipient, E_n, cpProof)` atomically; both verifications must succeed.

8 End-to-End Walkthrough

Alice owns a \$500K Calgary home (land \$200K, structure \$300K). Two insurers (LandCo, HomeCo) have issued BUCK_CREDIT NFTs; HomeCo’s structure credit depreciates LINEAR at 200bp/yr. After 3 years and partial activation, Alice’s collateral and effective coverage:

Token	Insurer	Face Value	Depreciation	Activated	Current Value	Premium Rate
#42	LandCo	200,000	NONE	200,000	200,000	50bp
#43	HomeCo	300,000	LINEAR, 200bp, 20% fl	150,000	141,000	200bp
Total					341,000	

8.1 Step 0: Identity Registration (One-Time)

Alice obtains a PS-signed credential off-chain from a trusted issuer (the Alberta government, ATB Financial, etc.), rerandomizes it, generates a fresh ElGamal key pair, encrypts her identity point $M = m \cdot G$ under her public key, and produces the Schnorr-family NIZK binding the rerandomized sigma to the ciphertext. `IdentityRegistry.register(issuer, pk, E, sigma', proof)` verifies both via BN254 precompiles in about 235K gas. After this call `isVerified[alice] = true` and Alice can touch BUCK. She repeats steps 2-3 for every Ethereum account she controls; the issuer never sees those accounts.

8.2 Step 1: Mint

Alice calls `BUCK.mint(50_000e6)` – the no-arg form, so the built-in selector sorts her NFTs by `premiumRate` ascending: LandCo (50bp) first, then HomeCo (200bp). At `BUCK_K = 1e18`:

1. `totalCurrentValue = 341,000e6, maxLimit = 341,000e6`. Alice's existing balance is 280,000e6, so the early gate ($280k + 50k \leq 341k$) passes.
2. The allocator walks NFT #42: `denom = 10000 - 50*10 = 9500, avail = 200,000e6 - mintsBacked[#42]`. Suppose this is Alice's first mint; `avail = 200,000e6, netCap = 190,000e6`. `netCap >= 50,000e6`, so the partial-fill closed form fires: `take = ceil(50_000e6 * 10000 / 9500) = 52,631,579, principal = take - 50_000e6 = 2,631,579`. `mintsBacked[#42] + 52,631,579 =`.
3. Tight gate: $280,000e6 + 52,631,579 \leq 341,000e6$ – passes.
4. Alice receives exactly 50,000e6. `insurancePool` receives 2,631,579 (the 10x principal whose 10% ROI funds the 0.50% annual premium on the activated coverage).
5. `Minted(alice, 52,631,579, 2,631,579, 341,000e6, 1e18, 341,000e6)`, plus standard `Transfer` events.

8.3 Step 2: Approve Bob (Identity Handshake)

Alice's wallet reads Bob's `pk_bob` from `IdentityRegistry`, decrypts `E_alice` (it knows `sk_alice`) to recover `M`, and re-encrypts under `pk_bob` with fresh randomness as `E_bob = (r_b \cdot G, r_b \cdot pk_bob + M)`. It produces a Chaum-Pedersen NIZK `pi_CP = (e, s_1, s_2)` proving `E_bob` encrypts the same `M` registered in `E_alice`, with Fiat-Shamir bound to `msg.sender = alice, spender = bob, chainid`.

`BUCK.approve(bob, 10_000e6, E_bob, pi_CP)` verifies both parties are `isVerified`, runs `IdentityRegistry`. (29K gas), stores `~_receiptFragments[alice][bob] = keccak256(E_bob)`, calls `markApproved(bob)` to freeze his `isCarrying` flag, and writes the ERC-20 allowance.

Off-chain (e.g. via Holochain), Alice delivers the matching `identity_data` and `E_bob` ciphertext to Bob's wallet. Bob decrypts to recover `M'` and asserts $H(\text{identity_data}) * G === M'$ – the two-channel binding from the Identity document.

8.4 Step 3: Bob Pulls the Transfer

`BUCK.transferFrom(alice, bob, 10_000e6)` debits the allowance, runs `_identityCheckedTransfer` (both parties verified, `~_receiptFragments[alice][bob] != 0`), and dispatches to the Non-Carrying path (Alice is not Carrying). Tokens move; `BuckTransferReceipt(alice, bob, 10_000e6, fromHash, toHash)` is emitted. An auditor with subpoena power over either Alice or Bob can recover the underlying identities; an outside observer learns only that addresses Alice and Bob exchanged 10,000e6.

8.5 Step 4: Alice Denominates as Notes

Alice picks flavour A2 and values [100e6, 25e6] (totalFace 125e6). Her wallet pads the live pair with 14 dummies (v=0) so the batch lands on the smallest pinned circuit (N=16), snapshots (oldRoot, nextLeafIndex) from chain, derives the depth-20 sibling paths in submission order, and runs `snarkjs groth16 fullProve` against `build/snark/mint_batch_n16/mint.wasm` and `mint_final.zkey`. The proof triple is abi-encoded into `proofBytes`.

On-chain:

1. `BUCK.approve(notes, 125e6, E_notes, pi_CP_notes)` – the always-CP rule applies even though Notes is bound under a Public Identity. The receipt fragment is openable on subpoena.
2. `Notes.mint(proofBytes, oldRoot, newRoot, nextLeafIndex, 125e6, [cm[0]..cm[15]])` – the adapter dispatches internally by `cms.length=16` to its pinned `MintBatch16` Groth16 verifier. On success, 125e6 BUCK moves from Alice to Notes, `roots[]` advances to the SNARK-attested `newRoot`, `nextLeafIndex` advances by 16, `noteFaceSum + 125e6=`. Alice keeps the live openings privately; the dummies are discarded.

8.6 Step 5: Spend a Note

Later, the holder of one of Alice’s notes (Alice herself, or a bearer she off-chain handed the opening to) redeems `cm[0]` for 100 BUCK to `recipient`:

1. The wallet reconstructs (`pathElements[20]`, `pathIndices[20]`) for `cm[0]` by replaying the `Minted` event log and pulling `cms[]` from each batch’s calldata.
2. `snarkjs groth16 fullProve` against `spend.wasm / spend_final.zkey` produces a proof binding [`currentRoot`, Poseidon-3(`rho`, `idHash`, 4242), 100e6, `recipient`, 1] to the witness.
3. `Notes.spend(proofBytes, currentRoot, nullifier, 100e6, recipient)` – the contract checks the root window, rejects a previously-seen nullifier, runs the adapter (~360K gas total), marks the nullifier burned, decrements `noteFaceSum`, and calls `buck.transfer(recipient, 100e6)`. Notes is bound Carrying, so the transfer dispatches to the carrying path automatically: the recipient receives the full 100e6 and absorbs an amount-weighted share of the pool’s average demurrage age via `buckSeconds`.

If HomeCo later reappraises the structure at \$280K, Alice’s effective collateral drops correspondingly. If her outstanding BUCK exceeds the new limit, the InsurancePool covers the gap on documented default events; the Jubilee accumulator is the long-horizon backstop.

9 Gas Cost Snapshot

Approximate gas costs from the current Foundry suite (Solidity 0.8.31, optimizer on). PID and oracle figures are estimates against unwritten oracle wiring.

Operation	Gas (approx)	At 30 gwei	Notes
BUCK_CREDIT.activate()	~50K	~\$4	Single SSTORE + even
BUCK_CREDIT.updateCredit()	~80K	~\$6	Multiple SSTOREs (in
IdentityRegistry.register()	~235K	~\$19	PS pairing + NIZK (o
IdentityRegistry.verifyApprove	~29K	~\$2	CP proof, per counterp
IdentityRegistry.verifySpendCP	~36K	~\$3	CP-DLEQ via EIP-196
BUCK.approve() (identity)	~97K	~\$8	29K CP + 46K allowa
BUCK.transfer / transferFrom	~80K	~\$6	Identity check + receip
BUCK.mint() (static BUCK_K, 1 NFT)	~150K	~\$12	NFT walk + allocator
BUCK.mint() (static BUCK_K, 2-NFT spillover)	~210K	~\$17	+ second NFT crystal
BUCK.burn() (1 NFT)	~120K	~\$10	Mirror of mint, no lim
BUCK.mint() (with PID update, 4-pool basket)	~450K	~\$36	+ 4 V3 TWAP reads -
BUCK.mint() (PID cached, within dT)	~155K	~\$12	one SLOAD on the ca
Notes.mint (mint_batch, N=1)	~461K	~\$37	Groth16 verify (~216K
Notes.mint (mint_batch, N=16)	~624K	~\$50	~39K/leaf – shipped s
Notes.mint (mint_batch, N=32)	~1.12M	~\$90	~35K/leaf (verifier ~4
Notes.spend (Groth16, B-shape)	~360K	~\$29	Groth16 verify + Carr
Notes.spendACP (Groth16 + CP-DLEQ, A-shape)	~410K	~\$33	+ verifySpendCP (~36
BUCK_CREDIT.currentValue / BUCK_K.currentBuckK	0	Free	View

The PID update cost is amortised across all mints within a `dT` window (60 s default; only the first mint after each window pays the oracle-read + PID-math cost; every other mint sees the cached one-SLOAD path). `Buck.burn` calls `compute()` too, so burns share the amortisation. The same contracts deploy on EVM L2s (Arbitrum, Optimism, Base) where gas is 10-100x cheaper.

10 Security Considerations

10.1 Oracle Manipulation

BUCK/USDT and basket pool prices use Uniswap V3 TWAPs (600 s shipped, governance-tunable per pool), so manipulation requires sustained capital commitment over the full window proportional to pool depth. The `BuckKControllerV3TwapTest` suite verifies that a single-block flash-shape attack driving BUCK to \$0.50 leaves spot collapsed but TWAP unchanged at ~\$1.000002, with the PID computing off the unchanged TWAP. When/if Chainlink basket feeds are adopted later, the Chainlink branch of `_getBasketCost` must verify `updatedAt` is within an acceptable window (~1 hour) and revert if stale.

10.2 Credit Enumeration DoS

A malicious user could mint many tiny BUCK_CREDITS to make `totalCurrentValue` and the cheapest-first allocator gas-prohibitive. Mitigations: minimum face value at credit creation, a cap on credits per account (e.g. 20), or off-chain aggregation with Merkle proof. None are wired in yet – the relevant test suites cap NFT counts at small N.

10.3 Insurer Trust

The insurer can reduce `faceValue` at any time, potentially pushing accounts into default. This is by design – it reflects real-world reappraisal risk. The `InsurancePool` covers default events; governance

can blacklist abusive insurers; multi-insurer market discipline is the natural check. Time-locks on insurer updates (e.g. 7-day delay for reductions > 20%) are a future enhancement.

10.4 Reentrancy

All state changes occur before external calls in `BUCK.mint`, `BUCK.burn`, and `Notes.mint / spend / spendACP`. Identity verification is the first guard in every transfer; `mintsBacked` is written before the user-facing `_mint`. `BuckCredit.activate` is a leaf operation. No `ReentrancyGuard` is currently wired – the design relies on checks-effects-interactions discipline; adding a guard before mainnet is a one-line change if review insists.

10.5 Pool Solvency

`insurancePool` holds the cumulative pool principal for every outstanding mint. Burn requires the pool's spendable balance to cover the proportional refund; if the pool is drained out-of-band the burn reverts. Today no path drains the pool: it is not a verified identity (so no identity-checked transfer can move BUCK out of it) and holds no NFTs (so it cannot self-burn via the allocator). Adding governance-controlled disbursement requires explicit accounting of how the pool's invariant (`poolBalance >= sum of all outstanding mint principals`) is preserved.

11 Future Directions

This implementation prioritises clarity and correctness; alternatives the architecture document flags for later study:

- **ERC-1155 batch credits** if standardised credit *classes* emerge (e.g. "Calgary residential, HomeCo, LINEAR 200bp"), enabling `safeBatchTransferFrom` for portfolio management and reduced gas for accounts holding many same-class credits. Per-token depreciation start dates partially negate the fungibility benefit, so applicability depends on the secondary market.
- **ERC-3643 (T-REX)** if regulatory KYC compliance becomes a deployment requirement. Native ONCHAINID, programmable transfer restrictions, agent-role granularity, and lost-key recovery via identity verification. This is the likely production standard if Alberta deploys BUCK under provincial regulation.
- **ERC-4626 InsurancePool** – a vault-shaped wrapping of the existing `insurancePool` address, so pool deposits become standardised mutual-insurance shares with `deposit / shares / yield` composability. Mechanical layer on top of the current single-address pool.
- **L2 deployment**. The contracts are EVM-compatible and deploy on Arbitrum, Optimism, Base, or Polygon with no changes. Cross-chain BUCK via canonical bridges or LayerZero/Axelar would enable multi-chain liquidity while keeping a single `BUCK_K` oracle on the settlement layer.
- **Coverage-ratio premium**. The current per-NFT `premiumRate` is flat. A loss-layer-style effective-rate derivation – $\text{rate}(V, A, D) = \lambda * \text{coverage} * (1 - D/V)^2 / 2$ under uniform loss – would price closer-to-full-coverage activations more aggressively, which matches actuarial intuition for parametric insurance with deductibles. The allocator's sort key would move from raw `premiumRate` to a midpoint-evaluated effective rate; everything else carries through.

- **Recursive SNARKs above N=512.** Mint batches above N=64 hit the EIP-170 24 KB bytecode ceiling on stock Groth16 verifiers. Calldata-IC delivery (hash-pinned) buys headroom to N=128 or 256; Nova / Halo2 / Plonky3 recursion is the natural fix above N=512.

12 Status

Current Foundry suite: 272 unit tests pass across 17 suites; one `BuckKControllerForkTest` requires a live mainnet RPC and is skipped offline. The shipped surface covers all five contracts in tree (`BuckCredit`, `IdentityRegistry`, `Buck`, `BuckKControllerStatic` + `BuckKController`, `Notes`) plus the SNARK toolchain (`mint_batch.circom` at pinned $N \in \{1, 2, 4, 8, 16, 32\}$, `spend.circom`, `spend_a.circom` V2) and a Python reference wallet (`alberta_buck.wallet`) that emits canonical test vectors the Solidity tests load via `vm.readFile`.

The BUCK_K test suites cover:

- `BuckKControllerUnitTest` – harness with mocked Chainlink feeds, mocked BUCK price. Includes priming, dS compensation, dTMax clamp, anti-windup, and multi-step integral convergence.
- `BuckKControllerV3Test` – locally deployed Uniswap V3 factory, all four basket pools at reference prices, BUCK/USDT pool with a 1e17 full-range LP (~100 K BUCK / 100 K USDT, matching the design narrative). Spot-price reads, swap-driven price drift, and PID response.
- `BuckKControllerV3TwapTest` – same V3 setup with `twapInterval` = 600 s on every pool, plus a warmup loop that walks forward writing observations on each pool. Verifies the TWAP path smooths a flash-shape spike (spot \$0.50, TWAP \$1.000002) and tracks sustained drift over the 600-s window (TWAP converges from \$1.00 → \$0.97 → \$0.95 over a full window at \$0.95 spot).

Outstanding work tracked in companion memos:

- **Mainnet-fork BUCK_K test** – exercise `BuckKController` against the four real on-chain pools at a pinned recent block, verifying basket cost reads sensibly against live data. Cheap validation step before mainnet.
- **Multi-block historical replay** – a backtest harness that walks `BuckKController` through 12 months of EVM history at a chosen block cadence, for tuning $K_p/K_i/K_d$ against a real price tape. Highest-value validation; biggest engineering lift.
- **PID-driven arbitrage scenario test** – the worked example in `alberta-buck-ethereum-example.org` demands a scripted Foundry test that drives Bobs through mint/swap/burn cycles while an Alice arbitrages around the PID equilibrium, with a Python visualisation of the trajectories.
- **Serialised B1 sub-notes** (`spend_serialized.circom` + per-parent bitmap) – design validated end-to-end by `scripts/snark/serialized_notes_poc.py` at $N_{fam} \in \{256, 1024, 4096\}$; circuit port and contract bitmap support are the implementation gap. See `alberta-buck-notes-serialized.org`.
- **Production trusted-setup ceremony** – replace dev entropy in `scripts/snark/setup.sh` with a multi-party ceremony per pinned circuit before mainnet.

The cryptographic surface is described by an executable Python reference; CI regenerates the canonical JSON test vectors and `git diff --exit-code`'s them against committed copies, so any drift between the spec, the wallet, and the Solidity tells us immediately which of the three is wrong. No `vm.ffi` shells out to Python at test time – the wallet emits vectors once, the Solidity reads them many times.